STRATEGY FOR NEW INDIA @ 75

MODULE - 12

12 ENERGY

OBJECTIVES

Emission Intensity of Economy (GDP) is a measure of greenhouse gas emissions per unit of economic activity, usually measured in GDP.

- It is to provide access to affordable, reliable, sustainable and modern energy.
- The following milestones are to be achieved
  - Make available 24x7 power to all by 2019.
  - Achieve 175 GW of renewable energy generation capacity by 2022.
  - Reduce imports of oil and gas by 10% by 2022-23.
  - Continue to reduce emission intensity of GDP in a manner that will help India achieve intended Nationally Determined Contribution target of 2030.

12 ENERGY

PRESENT SCENARIO

INDIA's ENERGY MIX

Coal 49.6%, oil 28%, biomass 11.6%, gas 7.3%, renewable and clean energy 2.2%, nuclear energy 1.2%

Per capita energy consumption is about 626 kg of oil equivalent against the world average of 1,860 kgoe. US and China's per capita energy consumption is 6,800 kgoe and 2,170 kgoe respectively.

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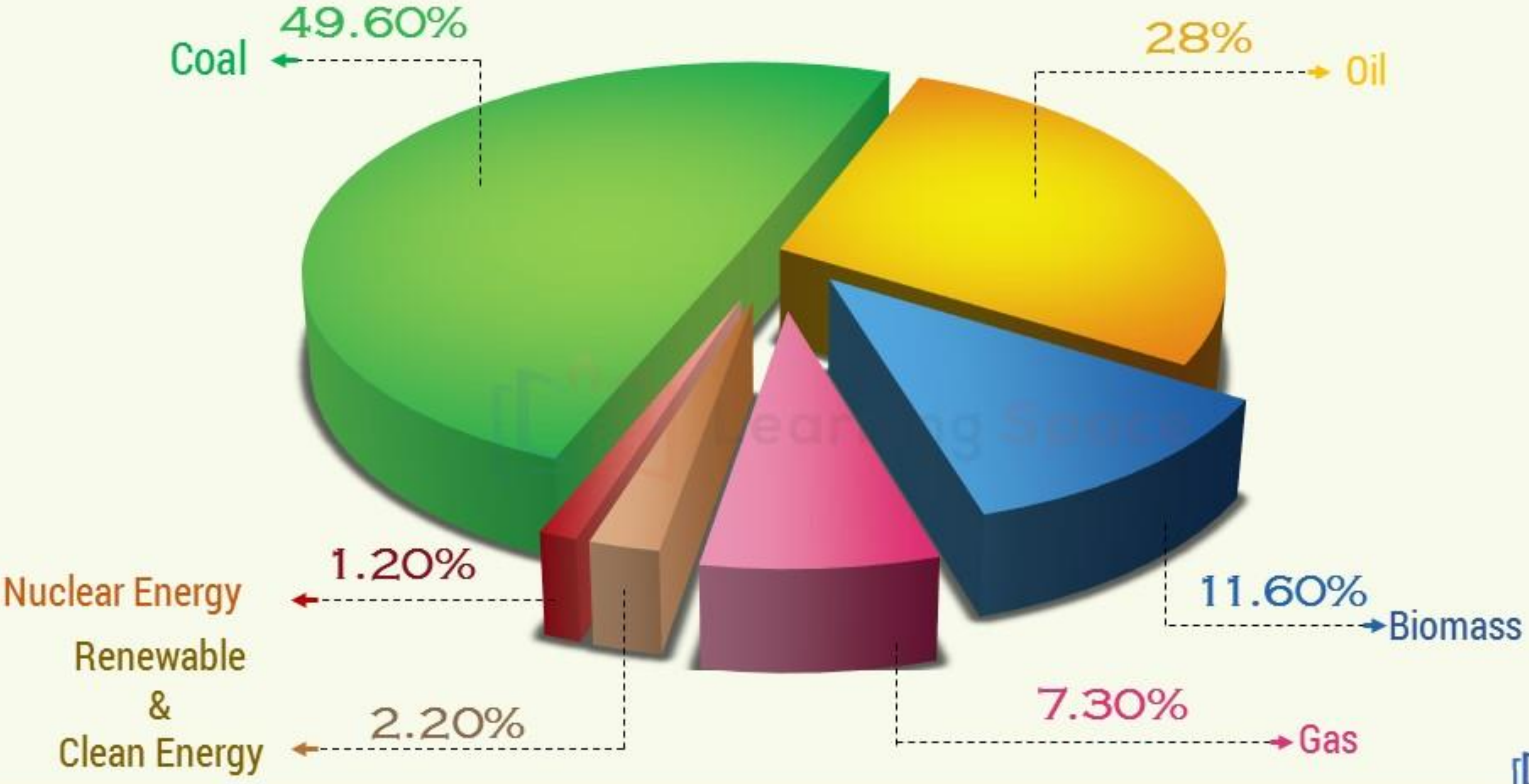

12 ENERGY

PRESENT SCENARIO

POWER CAPACITY

PNGRB

- All India installed power capacity is about 334 GW, including 62 GW of renewable energy.
- On energy supply, India is still heavily dependent on petroleum imports to meet its requirements.
- We imported approximately 82% of crude oil and 45% of natural gas requirements during 2017.
- At present about 16,000 km of gas pipeline network exist.
- The progress of 10,000 km of pipeline bid out by Petroleum and Natural Gas Regulatory Board, which was to be completed by 2016-17 is behind schedule.

12 ENERGY

PRESENT SCENARIO

ENERGY EFFICIENCY

- Due to the energy efficiency measures, India's energy intensity declined from 0.158 kgoe/$ in 2005 to 0.122 kgoe/$ in 2016 at 2005 prices.
- It indicates efficiency increase of 22.8%.
- The energy intensity of UK and Germany were 0.074 kgoe/$ and 0.101 kgoe/$, which indicates India still has the potential to improve energy efficiency.

12 ENERGY

CONSTRAINTS

OVERALL ENERGY

1. A variety of subsidies and taxes distort the energy market and promote the use of inefficient over efficient fuels.
2. They also make Indian exports and domestic production uncompetitive.
3. Energy taxes are not under GST. No input tax credit is given. This is the serious lacuna.

12 ENERGY

CONSTRAINTS

POWER SECTOR

1. Old inefficient plants continue to operate and more efficient plants are under utilized.
2. Gap between the Average Cost of Supply (ACS) and Average Revenue Realized (ARR) persists due to the high Aggregate Technical and Commercial (AT&C) losses.
3. DISCOMS resort to load shedding to minimize the losses.
4. Regulatory Commissions are unable to fully regulate discoms and fix rational tariffs though they are legally independent.
5. Unmetered power supply to agriculture provides no incentive to farmers to use electricity efficiently.
6. Lot of hidden demand because of unreliable supply and load shedding.
7. State power utilities are not able to invest in system improvements.
8. High industrial / commercial tariff and the cross-subsidy affected the competitiveness of the industrial and commercial sectors.

12 ENERGY

CONSTRAINTS

OIL & GAS

1. Non-discriminatory access for private and public sector companies in the gas pipeline network does not exist.

2. Lack of market-driven gas prices for old fields disincentivizes further production.

3. Gas pipeline infrastructure is inadequate.

12 ENERGY

CONSTRAINTS

COAL

1. Land for coal mining is becoming a contentious issue.

2. There is a tendency to expand opencast mining and discourage underground operation even for better quality coal reserves.

3. This aggravates land availability problem.

4. There is no competitive coal market.

12 ENERGY

CONSTRAINTS

RENEWABLE ENERGY

1. High energy costs result in reneging old power purchase agreements and erode their sanctity.

2. This leads to uncertainty regarding power offtake and consequently endangers further investments.

3. Flexibility in generation and balance requirements for the integration of renewable energy are emerging as major issues.

4. There are supply chain issues in biomass power generation.

DRIVERS

नीति आयोग

STRATEGY FOR NEW INDIA @ 75

Learning Space

12 ENERGY

CONSTRAINTS

ENERGY EFFICIENCY

1. Limited technical capabilities, high initial capital expenditure, limited market and policy issues have adversely affected energy efficiency.

2. High transaction costs which involves appointing suitable consultants and vendors for execution relative to project size, especially in MSME sector, makes energy efficiency investments unattractive for investors.

3. Non-availability of sufficient credit facilities and difficulties in obtaining required finances are strong deterrents to investments in energy efficiency in India.

सत्यमेव जयते
GOVERNMENT OF INDIA
MSME
MINISTRY OF MICRO, SMALL & MEDIUM ENTERPRISES

12 ENERGY

WAY FORWARD

1 OVERALL ENERGY

- Oil, natural gas, electricity and coal may be brought under GST to enable input tax credit.
- Have same GST rate for all forms of energy to enable a level playing field.
- All form of subsidies should be provided as functional subsidies to end-consumers to empower them to choose the energy form most suitable and economical to them.

12 ENERGY

WAY FORWARD

2 POWER

A smart meter is an electronic device that records consumption of electric energy and communicates the information to the electricity supplier for monitoring and billing.

- Promote smart grid and smart meters.
- All PPAs, including those with state generation companies should be based on competitive bidding.
- Privatize state distribution utilities and the use of a franchisee model will reduce AT&C losses.
- Regulatory bodies need to be further strengthened and truly made independent.
- For agriculture look at upfront subsidy per acre of land through Direct Benefit Transfer, instead of providing separate subsidies for fertilizers, electricity, crop insurance etc.
- Promote the use of solar pumps for agriculture.

नीति आयोग

12 ENERGY

WAY FORWARD

2 POWER ... *Contd*

- Local DISCOMs should buy surplus power from the farmer.
- DISCOMs may be fined for load shedding.
- Ensure effective enforcement of cap on cross subsidy.
- Actively promote cross-border electricity trade to utilize existing / upcoming generation assets.
- Introduce time-of-day tariff to promote the use of renewable energy.
- Introduce performance-based incentives in the tariff structure.

To manage the demand for power, it is necessary to introduce 100% metering, net metering, smart meters, and metering of electricity supplied to agriculture.

12 ENERGY

WAY FORWARD

3 OIL & GAS

ओएनजीसी
ONGC

ऑयल इंडिया लिमिटेड
Oil India Limited

- Provide a common carrier and open access to gas pipelines.
- Separate the developmental and regulatory functions of PNGRB.
- Expedite establishment of National Gas Grid.
- Promote city gas distribution to provide Piped Natural Gas.
- Ensure required flexibility in contract terms to make stranded oil and gas assets functional.
- Enhance production from the existing fields of ONGC and OIL using cutting-edge technology.
- Consider market pricing for blocks that are not viable because of low gas pricing.

12 ENERGY

WAY FORWARD

3 OIL & GAS ... Contd

- Provide "priority sector" status for 2G bioethanol projects.
- The concept of 'solar parks' can be applied to bio-fuels.
- Land can be leased by the government to Oil Marketing Companies (OMCs) for energy crops.
- Provide shared infrastructure for evacuation of oil & gas from small and scattered on-shore and off-shore fields.
- Promote PNG in rural areas.
- Create strategic reserves through various policy options.

12 ENERGY

WAY FORWARD

4 COAL

- Expeditiously complete detailed exploration through exploration-cum-mining leases based on production / revenue sharing model.
- State Governments should take the responsibility for making the land required for mining available.
- Expeditiously operationalize commercial coal mining.

12 ENERGY

WAY FORWARD

5 RENEWABLE ENERGY

- Provide a mechanism for cost-effective power grid balancing based on gas-based / hydro or storage.
- Renewable Purchase Obligations (RPO) should be strictly enforced and inter-state sale of renewable energy should be facilitated.
- It is necessary to have national level markets and regulations for balancing of power.
- Decentralized renewable energy in rural areas in conjunction with the DISCOMs grid can offer reliability.
- Hybrid renewable energy systems such as solar PV + biomass should be explored.
- Commercial biogas needs to be promoted by providing subsidy to consumers.

12 ENERGY

WAY FORWARD

6 ENERGY EFFICIENCY

- BEE should come out with a white paper on its 5-year strategy on energy efficiency and specify energy consumption norms.
- State designated agencies should be more empowered and provided with adequate resources to implement EE related activities.
- States should adopt second version of the Energy Conservation Building Code in their building by-laws.
- Promote the mandatory use of LED and the replacement of old appliances in Government buildings with five-star appliances.
- Focus UJALA Programme on lower-income households and small commercial establishments.

12 ENERGY

WAY FORWARD

6 ENERGY EFFICIENCY ... *Contd*

- The number of appliances covered under Standards and Labelling Programme should be increased.
- Widen and deepen the Perform, Achieve and Trade (PAT) Programme.
- Make Energy Saving Certificate (ESCert) trading under the PAT Scheme effective by ensuring strict penalties against defaulters.
- For the MSME sector, BEE should develop cluster-specific programmes for energy intensive industries to introduce energy efficient technologies.
- Promote the use of the public transport system.
- Expand the corporate average fuel efficiency standards beyond passenger cars to other vehicle segments.

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For Suggestions:
suggestions@learningspace.in

To Contact us:
info@learningspace.in

OUR TEAM

G. V. Rao	Amrita Naidu
K. Srikanth	M. Binukrishna
D. Sunil Kumar	G. Ravi Babu
L. V. Krishna	D. Joji
Y. Deepthi	K. Victor Babu

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